

Lionael Surya Dwitama

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PROFESSIONAL SUMMARY

Data Science and Analytics student at Seneca Polytechnic (GPA 3.9/4.0) who takes analysis from raw public data to a dashboard someone can audit. Built a validated nutrition-priority index for Indonesia's 38 provinces with spatial analysis and a live Streamlit dashboard, plus a tested dbt/PostgreSQL mart whose data-quality checks caught a KPI error before it reached a report. Works in Python, SQL, and R, and documents assumptions and data gaps as part of every deliverable.

EDUCATION

Seneca Polytechnic Toronto, ON
Honours Bachelor of Data Science and Analytics Sep 2024 – Apr 2028 (Expected)

- **GPA 3.9/4.0**; President's Honour List (Winter 2026); SSF International Student Merit Award
- **Coursework:** Statistical Methods, Data Preparation, Data Mining, Visualization, Business Intelligence, Data Privacy & Ethics

TECHNICAL SKILLS

Languages: Python, SQL, R, TypeScript
Analysis & Statistics: pandas, NumPy, correlation, ANOVA, PCA, sensitivity analysis, GeoPandas, PySAL (Moran's I, LISA)
Visualization & BI: Streamlit, Power BI, Tableau, Leaflet.js, Excel, Google Workspace
Data & Tooling: PostgreSQL, MySQL, dbt, Docker, Git/GitHub Actions (CI), pytest
Machine Learning: XGBoost, scikit-learn, SHAP, Optuna

PROJECTS

Nutrition Priority Index (Indonesia) | *Python, GeoPandas, PySAL, Streamlit, pytest* [GitHub](#) · [Live dashboard](#)

- Built a decision-support index ranking Indonesia's 38 provinces for child-nutrition intervention using only real government data from the BPS (Statistics Indonesia) WebAPI and Ministry of Health stunting records.
- Wrote the ingestion and cleaning pipeline with YAML data contracts, checksummed provenance logs, and name standardization; documented real coverage gaps (4 provinces missing from two BPS series) rather than imputing them.
- Validated before publishing: Spearman $r = 0.74$ against stunting outcomes (0.67 excluding partial-coverage provinces, both reported), ranks stable at $r > 0.99$ under $\pm 20\%$ weight changes, and PCA tested against a simpler single indicator.
- Ran spatial analysis (Moran's $I = 0.622$, $p = 0.001$; LISA) confirming a $2.5\times$ East-West gap via ANOVA and Kruskal-Wallis; shipped a 6-page Streamlit dashboard with maps, a filterable ranking explorer, CSV export, and a methodology page.

E-Commerce Analytics Mart | *dbt, PostgreSQL, SQL, Docker, GitHub Actions* [GitHub](#)

- Modeled the real Olist dataset (100k+ order rows) into staging, intermediate, and mart layers, including a monthly revenue KPI table, with one shared revenue definition.
- Caught a grain error that passed every generic test: the source issues a new customer ID per order, which would have shown every customer with exactly one lifetime order; rebuilt the customer dimension on the person-level ID.
- Wrote data-quality tests (not-null, uniqueness, relationships, accepted ranges, revenue reconciliation); a payment check flags 293 of 95,420 customers and is kept visible as a documented warning rather than suppressed.

FloodCast Jakarta | *XGBoost, SHAP, Optuna, Leaflet.js* Dicoding AI Impact Challenge 2026

- Built a flood-risk early-warning model for 15 Jakarta neighbourhoods using calibrated multi-horizon XGBoost tuned with Optuna, with SHAP explanations written in Bahasa Indonesia and shown on an offline map dashboard.

EXPERIENCE

Data Entry and Web Development, QuickRN Canada Jul 2025 – Present

- Built Aspen OS, an internal task, kanban, and documentation platform (Next.js, TypeScript, Supabase) the team uses daily.
- Maintain exam-prep content data and WordPress sites; troubleshoot course-access issues and document resolutions.

Career Peer, Seneca Works, Seneca Polytechnic Sep 2026 – Present

- Help students and alumni with resumes and cover letters; support career workshops and keep records in SharePoint.

IT & Media Strategy Associate, PERMIKA Toronto (volunteer student organization) Aug 2025 – Sep 2026

- Built event-management and member-engagement tools; led the League of Toronto 2026 campaign with a team of ten.

Software Engineering Intern, CV. Veda Sakti Dharma, Jakarta Jun 2025 – Aug 2025

- Developed a Python app automating PPh 21 income-tax calculations and payslips; validated payroll data for reporting.